ARUSHA REGION
FORM TWO MOCK 2026
PHYSICS
MARKING SCHEME

1.

i	ii	iii	iv	v	vi	vii	viii	ix	x
C	A	D	A	A	B	B	C	B	C

01@ (10 marks)

2.

i	ii	iii	iv	v
F	C	E	B	D

01@ = (05 marks)

3. @ i/ Magnetization
→ Is the process of making magnetic material to aquire magnetic properties. or
→ Is the process of making a magnet. 01

ii/ Demagnetization
→ Is the process of destroying the magnetic properties of a magnet. Or
→ Is the process of destroying a magnet. 01

iii/ Magnetic field
→ Is the area around a magnet where magnetic materials can experience the force of attraction. 02

(b)
i/ ~~Attract~~ magnetic materials
ii/ Like poles repel while unlike poles attracts
iii/ Magnetic forces are strongest at the poles
iv/ Magnetic lines of forces runs from North pole to South pole.
Any three 01 @ = 03

(c)
i/ Bar magnet
ii/ Horse-Shoe magnet (U-Shape magnet.)
iii/ Disc magnet
03

02

4. (a) i/ It is used to design various structure like, ships and planes (01)

ii/ Used to determine density of unknown substance using a known density of another (01)

iii/ Used to select building materials (01)

(b) i/ Data given

Mass of object (m) = 0.5kg = 500g

Volume of water before immersing the object (V_0) = $55cm^3$

Volume of water after immersing the object (V_1) = $65cm^3$

Volume of the object (V) = Required.

From: $V = V_1 - V_0$ (01)

$V = 65cm^3 - 55cm^3$

$V = 10cm^3$ (01)

∴ Volume of the object (V) = $10cm^3$ or $0.00001m^3$

ii/ Density of the object (ρ)

From: $\rho = \frac{m}{V}$ (01)

~~$\rho = \frac{0.5kg}{}$~~

$\rho = \frac{500g}{10cm^3}$

$\rho = 50g/cm^3$ or ~~$50,000kg/m^3$~~ (01)

∴ Density of object is $50g/cm^3$ or $50,000kg/m^3$

4. (c) Data given

Radius of the steel ball = 8 cm³

Volume of the steel ball (V) = $\frac{4}{3}\pi r^3$

$V = \frac{4}{3} \times 3.14 \times (8\,cm)^3$ (01)

$V = 2143.57\,cm^3$

From:

$1\,cm^3 \rightarrow 0.01\,N$

$2143.57 \rightarrow x$ (01)

$x = 21.4357\,N$

∴ The upthrust (U) = 21.4357 N (01)

5. (a) Kinetic Energy is the energy possessed by the body due to it's motion while Potential energy is the energy possesed by a body due to it's position. (02)

5. (b) Data given

Mass (m) = 200g = 0.2kg

Maximum height (h_0) = 20m

Rebound height (h_1) = 5m

i/ P.E before it fall

$P.E = mgh$ (01/2)

$P.E = 0.2 \times 10 \times 20$

$P.E = 40\ J$ (01)

ii/ From,

Maximum P.E = Maximum K.E (01)

Also → Loss in P.E = Gain in K.E

K.E as it hit the concrete = Maximum P.E

$K.E = 40\ J$ (01)

iii/ Velocity (V)

From, $K.E = \frac{1}{2}mv^2$ (01/2)

$v^2 = \frac{2K.E}{m}$

$V = \sqrt{\frac{2K.E}{m}} = \sqrt{\frac{2 \times 40}{0.2}}$

$V = 20\ m/s$ (01)

05

5. (b) iv/ K.E as it rebound.

From;

Gain in K.E = Loss in P.E

P.E, at 20m (01/2)

$P.E_1 = mgh$

$= 0.2 \times 10 \times 20$

$= 40J$ (01)

$P.E_2$ at 5m

$P.E = mgh$

$= 0.2 \times 10 \times 5$

$P.E_2 = 10J$ (00½)

From;

Gain in K.E = Loss in P.E

$K.E = P.E_1 - P.E_2$ (01)

$= 40J - 10J$

$K.E = 30J$ (01)

6. (a) i/ Water wets glass but Mercury does not wets glass because water has a high adhesion force compared to mercury which has high Cohesion force. (02)

ii/ Mosquitos walk on the Surface of water because of the Surface tension of water which prevent them from sinking. (02)

(b) i/ Data given

Force (F_1) = 5N

Extension (e_1) = 0.5m

Force (F_2) = ?

Extension (e_2) = 0.03m

From, $F \propto e$

$F_1 = k e_1$ (01)

$k = F_1 / e_1$

$k = \frac{5N}{0.5m}$

$k = 10 N/m$ (01)

Now, $F_2 = k e_2$

$F_2 = 10 N/m \times 0.03m$

$F_2 = 0.3N$ (02)

The weight which will stretch it by 0.03m is 0.3N

07

6. (b) ii/ Spring Constant (K)

From, $F = ke$

$k = \frac{F}{e}$ (01)

$k = \frac{5N}{0.5m}$

$k = 10\ N/m$ (01)

7. (a) A tractor with wide tyres can not get stuck in muddy places Compared to a car with narrow tyres because wide tyres provide large Surface which result to low pressure, while narrow tyres provide small surface area which results to high pressure. (03)

(b) i/ Depth (h) (01)

ii/ Density of the liquid (ρ) (01)

iii/ Gravity (g) (01)

$P = \rho g h$

7 (c) Data given:

Force (F) = 2 N

Pressure (P) = 0.2 N/m²

Area (A) = required

From: $P = \frac{F}{A}$ (01)

$A = \frac{F}{P}$

$A = \frac{2\ N}{0.2\ N/m^2}$ (01)

$A = 10 m^2$

Area (A) = 10 m² (02)

8 (a) i/ Principal focus

→ Is the point at which all rays parallel and close to the ~~principle~~ principal axis appears to converge at after reflection. (01)

ii/ Centre of Curvature

→ Is the centre of the sphere from which ~~the~~ mirror is made. (01)

8. (b) Data given:

focal length (f) = 4 cm

Object size (h_o) = 1.5 cm

Object distance (u) = 12 cm

i/ Image distance (v) =

From; $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$ (01)

$\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$

$v = \frac{uf}{u-f}$ (01)

$v = \frac{12 \times 4}{(12-4)}$

$v = 6$ cm (02)

Position of the image is 6 cm

ii/ Image size (h_i)

From; $M = \frac{v}{u} = \frac{h_i}{h_o}$ (01)

$\frac{v}{u} = \frac{h_i}{h_o}$

$h_i = \frac{v}{u} \times h_o$

$h_i = \frac{6}{12} \times 1.5 = 0.75$ cm (02)

Image size (h_i) = 0.75 cm

8 (b) iii/ The image formed is an inverted image

01

9 (a) Uniform velocity

→ Is the Constant rate of change of displacement.

01

(b) From;

1st equation of linear motion;

$$v = u + at \quad \text{--(i)}$$

For motion under gravity $(a = \pm g)$

$$v = u \pm gt$$

01

2nd equation of linear motion

$$s = ut + \tfrac{1}{2}at^2 \quad \text{--(ii)}$$

For motion under gravity $(s = h, a = \pm g)$

$$h = ut \pm \tfrac{1}{2}gt^2$$

01

3rd equation of linear motion

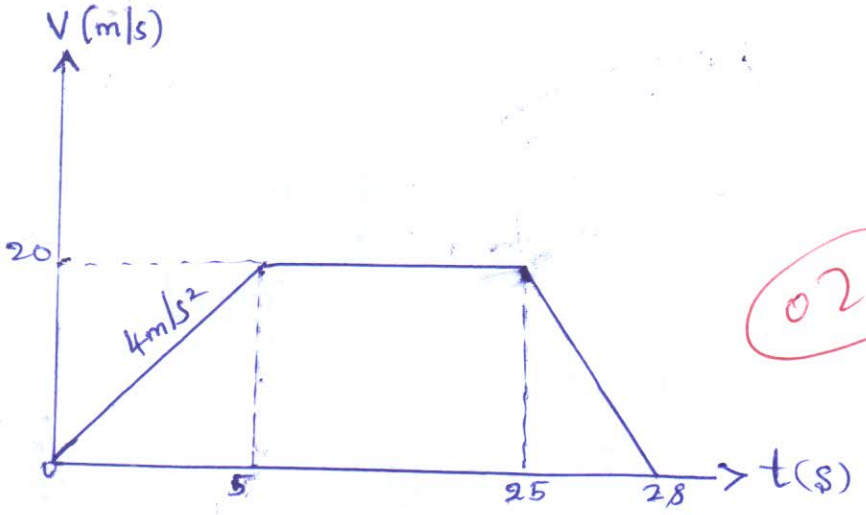
$$v^2 = u^2 + 2as \quad \text{--(iii)}$$

For motion under gravity $(s = h, a = \pm g)$

$$v^2 = u^2 \pm 2gh$$

01

9. (c) i/ The velocity-time graph for motion of the car.



02

ii/ Maximum velocity (V)

From: $V = u + at$

$V = 0\,m/s + 4\,m/s^2 \times 5\,s$

$V = 20\,m/s$ 02

iii/ Total distance Covered = Total area under the graph.

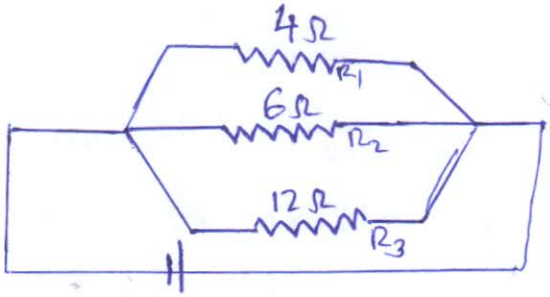
$= (\frac{1}{2} \times 5 \times 20) + (20 \times 20) + (\frac{1}{2} \times 20 \times 3)$

$= (50 + 400 + 30)\,m$

$= 480\,m$ 02

∴ Total distance = 480m

10. (a) i/



02

ii/ Total resistance (R_T)

From;

$$R_T = \frac{R_1 R_2 R_3}{(R_1 R_2) + (R_1 R_3) + (R_2 R_3)}$$ 01

$$R_T = \frac{4 \times 6 \times 12}{(4 \times 6) + (4 \times 12) + (6 \times 12)}$$ 01

$$R_T = \frac{288}{(24) + (48) + (72)}$$

$$R_T = 2\,\Omega$$ 01

(b) i/ Data given

Capacitance (C_1) = 3MF
Capacitance two (C_2) = 10MF
Capacitance three (C_3) = 15MF
Total Capacitance (C_T) = ?

$$C_T = C_1 + C_2 + C_3$$ 01

$$C_T = 3MF + 10MF + 15MF$$

$$C_T = 2MF$$ 01

10 (b) i/ The connection of capacitor in parallel will increases the ability of the capacitors to store charge. (01)

ii/ Lightning

→ Is a giant electric spark that arises due to discharge of atmospheric electricity. (01)

(c)

- Resistance box or Rheostat
- Voltimeter
- Ammeter
- Cell (01@ = 05)
- Connecting wires
- Resistor